

THERMAL And HUMIDITY EFFECTS On ELECTRONIC EQUIPMENT



Shavinder Singla is an IPC, USA certified interconnect designer, working as a technical officer at Centre for Development of Advanced Computing, Mohali

Apart from cabinet, an electronic assembly comprises electronic components assembled/soldered over a printed circuit board (PCB). When dealing with electronic assemblies, one must comply with certain standards with respect to its design, manufacturing and assembly. IPC, an association connecting electronics industries, is one of the organisations that develop such standards.

Based on performance, all the electronic equipment can be classified into three categories:

1. Class-1 general electronic products. These include consumer products, some computer and computer peripherals as well as general military hardware suitable for applications where cosmetic imperfections are not important and the major requirement is function of the completed printed board or printed board assembly.

2. Class-2 dedicated service electronic products. These include communication equipment, sophisticated business machines, instruments and military equipment in applications where high performance and extended life are required, and for which uninterrupted service is desired but is not critical. Certain cosmetic imperfections are allowed.

3. Class-3 high-reliability electronic products. These include commercial and military equipment where continued performance or performance on demand is critical. Equipment downtime cannot be tolerated, and equipment must function when required. Examples include life-support equipment and critical weapon systems.

These end-product classes have been established to reflect progressive increase in sophistication, functional performance requirements and testing/inspection frequency. There may be an overlap of equipment between classes. The printed board user needs to determine the class to which his product belongs. The contract shall specify the performance class required and indicate any exceptions to certain parameters, where appropriate.

Reliability

Reliability can have different meanings to different people. The IPC reliability series describes it as "the state where an equipment will work as long as expected, under a variety of use conditions, with no greater than a predicted number of failures." However, a general definition that has broad applicability is "time-dependent product performance that satisfies all customer expectations."

In military, industrial, commercial and consumer electronics applications, certain equipment may contain devices that are highly sensitive to environmental conditions. Understanding the effects of these uncontrolled environmental conditions at the component and system levels and applying this knowledge during the design phase can improve the reliability of the equipment, thereby reducing failures and thus maintenance costs.

Some of the environmental conditions affecting electronic equipment and systems include moisture, dust, vibration, air cooling and heating. The focus of this article is to quantify the effects of temperature and humidity on electronic components and PCB performance.

Thermal effect

As printed wire assemblies (PWAs) continue to increase in complexity, the risk of field failures due to unforeseen thermal problems also increases.

Integrated circuits. In the operation of an IC, electrons flow among tens of millions of transistors, consuming power. This produces heat, which radiates outward through the chip package from the surface of the die, increasing the IC's junction temperature. Exceeding the specified maximum junction temperature causes the chip to make errors in its calculations or perhaps fail completely.

When IC designers shrink a chip and reduce its operating voltage, they also reduce its power dissipation and thus heat. However, shrinking a chip also means

In order to design highly reliable products, designers need to study the intended application environment and its impact on the electronic assembly. This article discusses thermal and humidity effects and protection measures in particular